

Advanced Data Structures Lab Assignment 1

Problem Statement 1 (Part A):

Write a C Language program to implement the following functions for a binary search tree using linked list or arrays:

1. Search & Add elements in the binary search tree
2. Print the binary tree so that it looks like a tree (level wise printing).
3. Searches the maximum value in the tree.
4. Implement pre-order traversal and print the nodes of the tree using recursive and non-recursive methods.

Problem Statement 1 (Part b):

Write a C Language program to Implement the binary search tree for string values using linked list or arrays.

1. Search & Add string elements in the binary search tree
2. Print the binary tree so that it looks like a tree (level wise printing) of the strings values in the binary search tree.
3. Implement pre-order traversal and print the nodes of the tree using recursive and non-recursive methods.

Lab File Writing:

1. Algorithm to Search & Add nodes in a binary search tree.
2. Algorithm to search the maximum value in the tree
3. Algorithm for pre order traversal with & without recursion.

Node Structure using C language

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
// Define structure for a binary tree node
```

```
struct Node {
```

```
    int data;          // Value stored in the node
```

```
    struct Node* left;  // Pointer to left child
```

```
    struct Node* right; // Pointer to right child
```

```
};
```

```
// Function to create a new node
```

```
struct Node* createNode(int value)
```

```
{
```

```
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
```

```
    newNode->data = value;
```

```
    newNode->left = NULL;
```

```
    newNode->right = NULL;
```

```
    return newNode;
```

```
}
```